

Homework 6 Capstone

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Problem & Goals

- Build an AI-assisted transaction processing pipeline
 - Validate → detect fraud → compliance check
 - Demonstrate multi-agent workflow, skills, hooks, and MCP
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Architecture

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sample-transactions.json → Orchestrator
  → Validator → Fraud Detector → Compliance → shared/results/
Hono API ↔ Svelte Dashboard
MCP: context7 + pipeline-status
```

File-based JSON envelopes in `shared/` directories connect stages.

Pipeline Stages

Validation

- Required fields, positive amounts (decimal.js)
- ISO 4217 currencies: USD, EUR, GBP, JPY
- Rejects invalid records early

Fraud Detection

- Risk score: high-value (+40), cross-border (+25), unusual timing (+20), wire (+15)
- Threshold 50 → `fraud_review`

Compliance

- Flags high-value wires and fraud-review cases
 - Writes `pipeline-summary.json`
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Demo Flow

1. `npm run pipeline` — CLI processes 8 sample transactions
2. `npm run dev:api` + `npm run dev` — Svelte dashboard
3. MCP: `get_transaction_status` for TXN006 → rejected

Expected: 4 approved, 2 `fraud_review`, 2 rejected, 2 `compliance_flagged`

AI Workflow

Agent	Deliverable
Agent 1	specification.md, /write-spec skill
Agent 2	Pipeline, Hono API, Svelte UI
Agent 3	/run-pipeline, coverage gate hook
Agent 4	Vitest suite, README, presentation

context7 — Hono, decimal.js, Svelte lookups in research-notes.md

Testing & Quality

- **Vitest** — unit tests per stage + integration test
 - Isolated tmp dirs (no real shared/ pollution)
 - **Coverage gate** — hook blocks push if < 80%
 - Current coverage: ~87% statements, ~81% branches
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Lessons Learned

- Spec-first development reduced rework (TXN004 score edge case)
 - `decimal.js` avoids float bugs on monetary values
 - File-based IPC is simple to debug and test with tmp directories
 - MCP bridges AI chat with live pipeline state
 - Skills + hooks encode team conventions into the IDE
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Thank You

Repository: `homework-6/`

Docs: README.md, HOWTORUN.md, docs/MCP.md